

Histopathology

Localized scleroderma (LS) is primarily a clinical diagnosis, and skin biopsies for histopathological evaluation should be reserved for atypical or diagnostically uncertain cases. When performing an incisional biopsy, it is essential to obtain sufficient depth, as certain LS subtypes—such as eosinophilic fasciitis—may predominantly affect the subcutis, fascia, or even underlying muscle. In a study of 128 patients with eosinophilic fasciitis, eosinophilia and fibrosis were identified as predictive factors for disease relapse, while oedema, relapse, and fibrosis were associated with the development of sequelae. Histopathologically, it is not possible to reliably distinguish between LS and systemic sclerosis (SSc), nor to differentiate among the various subtypes of LS.

Laboratory Parameters

There are no specific serological markers for LS. However, routine laboratory tests should be performed—particularly before initiating systemic therapy—and should include complete blood count with differential, clinical chemistry panel, erythrocyte sedimentation rate (ESR), C-reactive protein (CRP), and antinuclear antibody (ANA) testing. Abnormal laboratory findings are common, especially in juvenile LS, with ANAs detected in up to 40% of patients. The presence of ANAs is associated with an increased risk of extracutaneous manifestations (e.g., arthritis) and higher likelihood of disease relapse. During the active phase of generalized LS, peripheral blood eosinophilia may be observed. In patients with linear LS affecting the extremities and concomitant joint involvement, elevated rheumatoid factor levels may be present and can correlate with the severity of arthritis. Additional immunological testing (e.g., screening for antibodies against extractable nuclear antigens) should only be conducted if there is clinical suspicion of systemic sclerosis. Serological testing for *Borrelia burgdorferi* is not routinely recommended in LS and should be limited to cases with clinical features suggestive of Lyme disease.

Imaging

Neurological involvement is reported in up to 50% of patients with head or facial LS (e.g., 'en coup de sabre' or progressive facial hemiatrophy), with symptoms such as migraine, headaches, and, less commonly, epilepsy. Cranial magnetic resonance imaging (MRI) is recommended to evaluate for subcortical calcifications, brain atrophy, or other structural abnormalities. However, many patients remain asymptomatic despite radiological findings. Therefore, baseline brain MRI is advised in all cases of head/face LS, or performed during follow-up if new neurological symptoms arise.

Referral to ophthalmology or oral surgery may be necessary depending on disease location and progression. MRI and computed tomography (CT) can also aid in surgical planning—particularly in 'en coup de sabre' type LS—and help assess involvement of muscle, joints, or bone, especially in linear LS of the extremities.

Instrument-Based Outcome Measures

Several instrument-based methods have been used in clinical trials on LS, including ultrasound, cutometry, durometry, thermography, laser Doppler flowmetry, and computerized skin scoring systems. These tools are typically employed as secondary outcome measures. Standardized photographic documentation of skin lesions is strongly recommended to support clinical assessment and monitor disease progression or response to treatment.

Differential Diagnoses

A broad differential diagnosis must be considered in the evaluation of LS, depending on the subtype and disease stage (see Table 2). Early diagnosis is crucial, as delayed recognition is linked to prolonged disease activity and increased relapse rates. Unlike systemic sclerosis (SSc), LS does not present with characteristic facial features such as telangiectasia, beaked nose, or microstomia, nor with vascular manifestations like Raynaud's phenomenon, pitting scars, or digital ulcers. Additionally, highly specific autoantibodies such

as anti-centromere and anti-Scl-70 (topoisomerase I) antibodies are absent in LS.